

Tablet as an enabler for Immersive Visceral Experiences

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Figure 1: User exploring extreme fire visualization system in the 360-degree cinema using a tablet

Abstract

This paper presents the preliminary results and lessons learned from developing a tablet as a touch-based interface within a 360-degree immersive environment to enhance firefighters' understanding of fire behavior. The tablet interface was developed through a User-Centered Design (UCD) and Design Thinking (DT) cycle to create a viscerally engaging experience. We evaluated the tablet with firefighter personnel. Our first results revealed that the tablet interface helped with situational awareness by providing context. Future efforts will focus on improving the UCD and DT cycles with more users, aiming to enhance the tablet to help develop immersive training modules that enhance firefighter preparedness and decision-making.

CCS Concepts

• Human-centered computing → User interface design.

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Keywords

Immersive Systems, Wildfires, Firefighters, Viscerality, Design Thinking, User-Centered Design, Interaction Design, UX Design

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1 Introduction

Extreme wildfires, characterized by their unpredictable and erratic behavior [39], are unplanned fires that affect vast open areas [1]. Their uncontrolled spread has led to an increase in mega-fires, which burn over 4,500 hectares of land [42]. These fires pose significant risks to flora, fauna, nearby populations, and the first response agencies confronting them [5]. Traditional first-response strategies, based on the assumption of linear wildfire evolution influenced by weather, topography, and vegetation [39]. However, these are proving insufficient as climate change accelerates wildfire speed and scale, challenging our understanding of these events [34, 44]. This unpredictability has shifted firefighting approaches from proactive suppression to more reactive strategies [5], highlighting the urgent need for advanced training tools to prepare first responders for the challenges posed by extreme wildfires.

To address these issues, firefighter associations in countries like Canada, the United States and Australia have developed guidelines like LACES (Lookouts, Awareness, Communications, Escape Routes, Safety Zones) [15]. However, teaching often relies on outdated

delivery methods like slide presentations, which fail to prepare firefighters to practice and apply LACES under the unpredictability of wildfires [5, 13, 19]. Firefighters need a visceral understanding of wildfire dynamics to build situational awareness and make effective decisions under pressure [32]. A deeper understanding of these practices can aid in preparedness, improve response efficiency, and reduce fatalities and psychological trauma [26].

Reviews from 2021 [22, 50] and 2023 [9] highlight situational awareness as a critical gap in firefighter teaching and training, emphasizing the need for visceral exposure to wildfires to understand their behavior better. A visceral understanding of how a fire spreads would make the delivery of LACES more valuable in the context of a wildfire. Immersive systems address this by enabling users to experience extreme events in controlled virtual environments safely [55]. Using technologies like projectors or head-mounted displays, these systems replicate realistic virtual worlds, offering a sense of being transported elsewhere [40]. Past efforts have developed immersive tools to prepare firefighters for various scenarios [9, 22, 50], allowing them to experience "unlikely" wildfire situations before encountering them in real life [4, 34, 51].

1.1 Touch User Interfaces and Viscerality in Wildfire Systems

Visceral spatial interaction involves instinctive, emotional responses and pre-reflexive reactions triggered by intuitive user-system interactions [7], achieved through sensory elements like visuals, touch, sound, and even scent [16]. While Norman's concept of emotional design [38] primarily focuses on immediate emotional responses to aesthetics, our approach extends this definition to encompass interfaces that could trigger reflexive responses and understanding through familiar interactions. Ghazali and Dix [16] highlight how interacting with familiar physical objects and spaces enhances the visceral feeling of presence. Combining these elements with hyper-realistic environments fosters familiarity in immersive spaces, which can be further explored to improve learning and situational awareness during extreme events.

Such experiences are powerful tools for understanding complex topics, as seen in *Data Visceralization* (DataVisc) proposed by Lee et al. [28], where immersing users in raw data enhances comprehension. This concept could be particularly relevant for wildfires, where *viscerality* fosters situational awareness in chaotic scenarios [6, 9, 23].

Beyond visual fidelity, true *viscerality* relies on intuitive and familiar interactions that evoke an instinctive connection [2]. For example, touchscreen interfaces can bridge physical and virtual worlds, map real-world inputs to digital outputs, and foster a sense of embodiment [25, 54]. This combination of familiarity and enhanced interaction deepens user engagement and facilitates learning in immersive environments [24].

In the context of firefighting training, multiple immersive systems have employed interactive touch-screen UIs, hoping to improve the user's situational awareness and the system's learning outcome [4, 20, 35, 52]. Previous projects showed that user interfaces based on familiar firefighting tools and immersive technologies improved system familiarity learning outcomes. Our project

builds on this by developing a new tablet-based interface to aid the understanding of wildfire behavior.

We guided our interface design by LACES [15] and developed a tool to serve as an anchor between the virtual environment and what firefighters learn at the academy, hoping to create a visceral engagement, that could improve situational awareness and decision-making teaching. Implementing a touchscreen-based interface, in this case, a tablet, into an immersive visualization system can bridge the gap between the digital and physical worlds, making interactions more intuitive and accessible [24, 25]. The tablet will allow users to manipulate digital information, providing a more natural and engaging user experience, creating a two-way communication channel with the system presented here.

1.2 Contribution

This work presents our initial steps toward developing a tablet-based interface for an immersive wildfire visualization system called *iFire*, to enhance firefighter training and decision-making. We conducted User-Centered Design and Design-Thinking cycles to create a prototype that aids the understanding of wildfire data and wildfire behavior. The goal was to achieve intuitive navigation and situational awareness inside the virtual environment. We built a tablet interface because it is familiar to users and has the potential to seamlessly bridge the gap between physical interaction and virtual environments. Through the tablet, we hope to design an interface that, in combination with *iFire*, supports situational awareness, visceral experience, and decision-making. This work marks the first step toward a more comprehensive training tool that aligns with firefighter needs and lays the foundation for future research.

2 Methods and Materials

2.1 Design Methods

We decided to use User-Centered Design (UCD) and Design Thinking (DT) because these methodologies complement each other in creating user-focused, practical solutions. UCD emphasizes understanding user needs, specifying requirements, and iteratively refining solutions to ensure usability [29]. It is particularly valuable for designing products with the end user's needs, preferences, and limitations in mind [27, 45, 49] while addressing the technical needs of users and determining how technology can meet those needs [37]. On the other hand, DT focuses on empathizing with users, defining problems, ideating solutions, prototyping, and testing [3]. Its strength lies in understanding the *emotions* and *feelings* of the users and how interactions with technology can shape their experiences [10, 41, 46, 48].

We combined these methodologies by leveraging UCD's structured, usability-driven approach alongside DT's focus on user insights and emotional engagement [12, 45, 46]. This project combines both methodologies to generate a robust design process that benefits from each methodology's strengths. By assessing firefighters' needs, we plan to develop a tool that addresses practical requirements that could possibly lead to a visceral experience.

2.2 User Interviews and Personas

In UCD and DT, interviews are essential for gathering insights and testing usability. UCD uses them in the "Understand the Context" and "Evaluate Designs" stages, while DT applies them in the "Empathize" and "Test" stages. For this stage, we conducted a semi-structured interview with an expert firefighter, chosen for their ability to uncover wildfire-fighting complexities and provide specific insights into training needs. Moreover, he has had an extensive and successful career as a wildfire dynamics researcher. These insights informed the design of an intuitive, interactive interface tailored to firefighters' challenges.

The Persona Method, a core element of UCD and DT, was also integral to our project as it addresses usability by representing users' tasks, experiences, and needs [18, 36]. This method allowed us to analyze user challenges and guide the interface design without directly involving all potential users. By creating three detailed personas, we ensured that our designs aligned with firefighters' goals and behaviors, fostering empathy and supporting a functional user-centered interface. These methods were chosen for their ability to align the design process with the specific demands of wildfire training.

2.3 Immersive System

iFire, which is our Immersive Visualization System comprises a back-end data modeling and processing pipeline, a front-end visualization engine, and a user-centered tablet controller.

Visualization Dataset: A professional fire analyst modeled the wildfire data for this study. The dataset represents a hypothetical fire in a pine plantation in South Australia. The fire was modeled to explain how multiple wind intensities, combined with various pine treatments, affect the spread of a wildfire. **Back-End System:** Our back-end system was built using the Python programming language. It can interface with the SPARK Wildfire Platform [21, 33], a professional platform for modeling wildfires. *iFire* can process the data so the front-end system can visualize it. **Visualization Engine:** Our visualization engine runs in a fully immersive, 360-degree cinema setup [11, 30]. The visualization front-end was built using Unreal Engine. Every variable from the modeled wildfire was represented as realistically as possible. The professional fire analyst visually inspected it to ensure it was as close to real life as possible. **Tablet Interface:** For the end user to interact with the fire, we built an interface within an Android tablet, also using Unreal Engine. This interface allows interaction with the immersive system and modifications of the scenario variables.

3 Design Cycle

3.1 Interview Responses and Persona Development

Our interview confirmed our findings about how unpredictability challenges emergency responders. The essence of contemporary firefighting proficiency lies in quickly interpreting unfolding situations and adapting to unforeseen environmental dynamics. Notably, the current training landscape lacks a comprehensive understanding of the intricate interplay of variables influencing fire behavior. This leads to firefighters struggling to make fast decisions when

unexpected events unfold. Another issue mentioned was the lack of emphasis on situational awareness training and the need to study fire progression from multiple angles to better understand its evolution.

Based on the expert's interview and other casual talks with firefighters from Fire and Rescue New South Wales (FRNSW), we developed three personas to explore their potential needs. We included goals and pain points (part of the Design Thinking cycle) into our personas to better understand their feelings during firefighting training. This is to design the features that will cover the needs of our final users. In-depth information about the personas can be found in Appendix A.

3.2 User Needs and Insights

The interview with our expert user and the personas we developed led to the following needs (an outcome of the UCD cycle) and insights (a result of the DT cycle). These will be used to create the initial prototype of our tablet's UI.

Needs: **[N1]Comprehensive Wildfire Understanding:** There is a need for tools that visualize and help understand the complex variables affecting wildfire behavior to enhance training and firefighting strategies. **[N2]Modernized Training Methods:** Firefighters require updated, realistic, and immersive training modules that reflect the unpredictable nature of wildfires and provide safe environments for learning. **[N3]Enhanced Decision-Making Skills:** Training systems must focus on helping firefighters quickly interpret and adapt to unfolding situations during wildfires, addressing the lack of comprehensive training in this area.

Insights: **[I1]Unpredictability of Wildfires:** Unlike conventional fires, wildfires introduce high unpredictability, challenging emergency responders and necessitating improved training methods. **[I2]Gap in Current Training:** Existing training methods do not adequately cover the intricate interplay of variables influencing fire behavior, leading to difficulties in fast decision-making during wildfire events. **[I3]Visceral Preparedness:** Experienced and aspiring firefighters need training that addresses their emotional and instinctual responses to effectively handle the stress so they can deal with the unpredictability of wildfire situations.

3.3 Basic Interaction

Once we identified key needs and insights from the firefighter interview and our personas, we defined a set of essential interactive features for the tablet interface. These features enable users to perform tasks and explore the immersive wildfire visualization system, addressing the challenges highlighted in our insights (refer to Section 3.2). This approach ties closely to the LACES [15] strategy by prioritizing situational awareness. The navigation system supports spatial and temporal exploration, empowering users to move through different locations dynamically and observe wildfire progression over time. Spatial navigation allows users to seamlessly explore diverse environments affected by wildfires, providing an immersive and hands-on experience of the complexities firefighters face on the ground. Temporal navigation adds another dimension by enabling users to travel through time and observe how wildfires evolve across periods. These features focus on understanding fire dynamics and progression when planning responses.

3.4 Interface Development

To experience the basic interaction we mentioned above and for the tablet to have a role as an enhancer of the overall experience, we established three interactive capabilities the tablet's UI needs to have. The UI will have slight modifications based on the current view.

(IC1): Perspective Switch. This feature addresses [N1] and [I2]. By allowing users to switch between *Aerial View* and *Ground to Crown View*, the system provides situational awareness from multiple angles essential for understanding the complex variables affecting fire behavior. The *Aerial View* offers a broader perspective from above to observe fire progression patterns and scale. It intends to help firefighters plan strategic responses and anticipate fire spread. A grid system will be used for better precision when switching views. Figure 2 B1 shows the aerial view with the grid overlay. The *Ground to Crown View* delivers an on-the-ground perspective (Figure 2 B2), highlighting fire interactions at surface and canopy levels (Figure 2 C1), enabling a deeper understanding of the fire's behavior. This aligns with the LACES [15] strategy by emphasizing situational awareness and continuous assessment of evolving fire conditions.

(IC2): Tablet as a Guide. The tablet's UI enables fire instructors to guide users through the virtual environment, simulating a "field trip" to a wildfire. This aligns with [N2] by creating a controlled yet immersive training module. Such guidance ensures that training remains focused, engaging, and aligned with the LACES [15] principles like communication and clear chain-of-command during wildfire scenarios.

(IC3): Tablet as an Aid. This feature addresses [N3] and [I3]. By displaying simplified wildfire data, such as fire arrival times or weather conditions (see Figure 2 D1), the tablet reduces visual clutter in the immersive environment. This facilitates quicker interpretation of unexpected wildfire behavior, helping users adapt in real-time—mirroring the adaptability required in real-world scenarios.

3.5 First Iteration Discussion

We are designing a tablet system to create a direct and intuitive connection between user input and *iFire*, enabling smooth engagement and real-time interaction. The tablet serves as both an instructional guide and a decision-making tool, focusing on situational awareness, communication, and evaluation. It aligns with LACES principles and addresses key gaps in firefighter training.

This system aims to enhance technical skills while preparing firefighters emotionally and instinctively for wildfires' unpredictability. The tablet's touchscreen lets users view and interact with wildfire data from *iFire*. Simplifying variables such as fire arrival time and weather conditions on the tablet can reduce visual clutter, which may otherwise overwhelm the virtual display.

An alternative display to view a simplified version of them can reduce the visual clutter, addressing IC3. Using a coordinated, tablet-based system has improved performance and user experience in 3D interaction tasks [47]. For details about the first UI iteration, see Appendix B.

3.6 Internal Workshop and Second Iteration

In an internal workshop, our development team identified usability issues with the tablet interface, leading to four key UI redesigns:

The *Utility Bay* (Figure 2, A) allows switching between **Aerial Mode View** and **Ground Mode View**, setting map interest points, and cycling through wildfire intensity presets (Moderate, High, Extreme). The *Global Scenario Information section* (Figure 2, B) displays the fire-affected area map, key wildfire variables (wind speed and direction, fire intensity, humidity), and view mode data, including area averages in **Aerial Mode View** and a 100m radius in **Ground Mode View**, time, date, season, and location coordinates. The *Elevation Visualization Tool* (Figure 2 C) enables y-axis camera movement, showing height in hundreds of meters in **Aerial Mode View** and relative to tree height and canopy fire intensity in **Ground Mode View**. The *Space and Time Navigation Module* (Figure 2 D) provides movement (forward, backward, left, and right) and time controls with a timeline of wildfire impact projections, allowing users to play, pause, bookmark, and explore simulation moments.

4 End-User Testing

We presented the second design iteration to firefighters from FRNSW in two workshops with four participants. After a brief introduction, we gave a 20-minute overview of the 360-degree cinema and demonstrated *iFire*. The 20-minute demonstration showcased the *Ground to Crown View* and *Aerial View*, fire spread dynamics, elevation and time navigation tools, and immersive wildfire navigation using the tablet controls. Firefighters then had 10 minutes of free exploration utilizing the tablet, followed by a 15-minute semi-structured interview, where they provided feedback, suggested improvements, proposed new features, and validated user needs.

4.1 Session feedback

Firefighters found the tablet intuitive and natural, particularly when moving around the 360-degree cinema. The elevation tool was highly valued as it provided rare views from tree canopies, offering a better understanding of erratic wildfire behavior, which often occurs at the crown level, compared to observing the spread solely from the ground.

The grid system and its movement feature were highly useful for firefighters, allowing seamless teleportation between **Aerial Mode View** and **Ground Mode View** and a clearer sense of location. It was mentioned that the tablet's mini-map provided a broader view of the ongoing events in the virtual environment. Additionally, wildfire variables such as temperature, wind speed, and smoke density displayed on the tablet screen were invaluable for interpreting fire behavior, particularly in low-visibility conditions caused by thick smoke in the scenario.

Firefighters appreciated the time manipulation controls for understanding fire spread over varying periods. They suggested adding timeline markers to revisit key fire events for further study. Improvements included simplifying teleportation in **Ground Mode View**, which currently requires switching to **Aerial Mode View** and back, a process that was found to be cumbersome. They also noted visual discomfort caused by the virtual joystick while

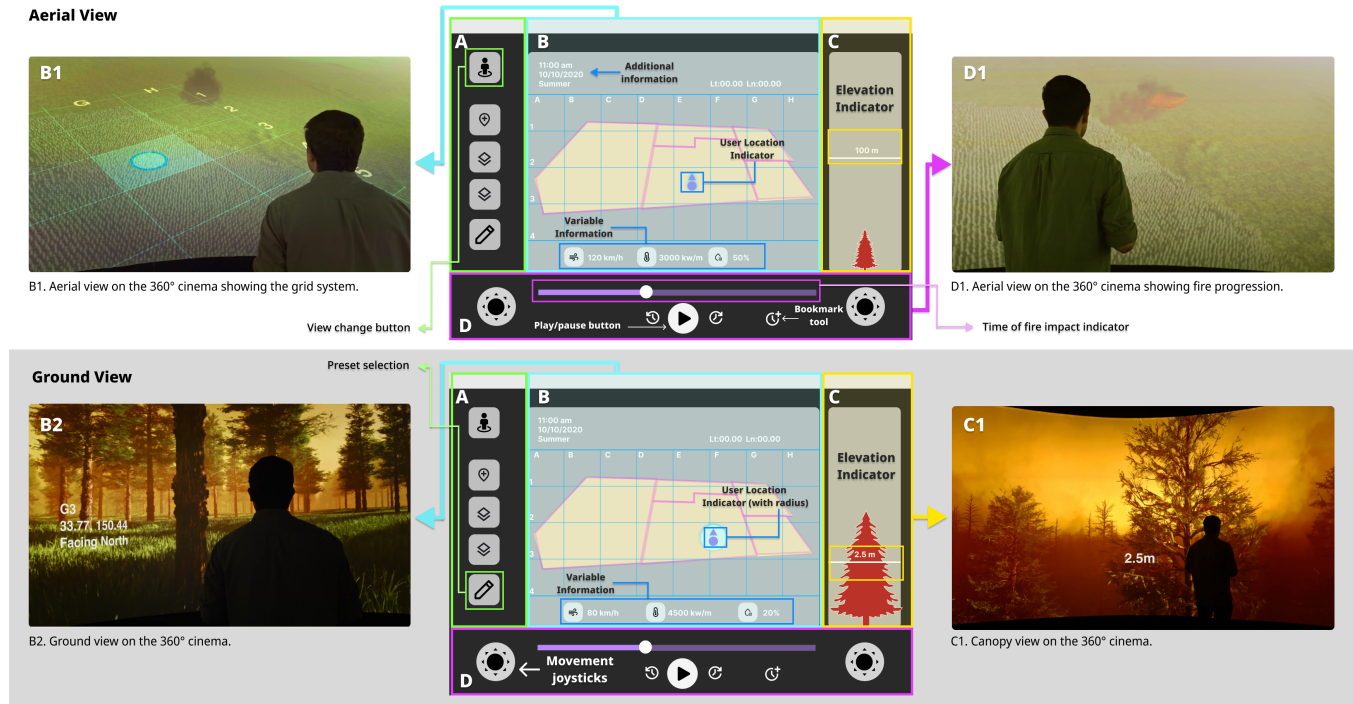


Figure 2: Diagram showcasing both aerial and ground view of the 360° cinema and both versions of the tablet's UI

turning around in **Ground Mode View**, as the 360-degree rotation was disorienting.

5 Discussion

5.1 Understanding Firefighters

During the discovery phase of the UCD and DT cycles, we observed a potential gap in training: limited understanding of variables influencing fire behavior, which is essential for effective firefighting. Reviews [8, 9, 14, 22, 50, 51], and our initial user interview suggests that situational awareness remains a challenge, as observing fire progression from multiple angles is not feasible during real-world training. Our canopy view feature aims to address this by offering a perspective that could provide firefighters with valuable insights typically unavailable in the field.

Preliminary findings from the persona method and firefighter workshops indicate the need for tools to visualize complex wildfire variables, immersive training to reflect wildfire unpredictability, and systems that enhance decision-making under stress. These observations align with the LACES strategy by emphasizing situational awareness and communication. This initial exploration offers a starting point to refine these features further and assess how they might support LACES-based training.

5.2 Tablet for a Visceral Immersive Exploration

During the prototype stage of our UCD and DT cycles, we developed a tablet-based touchscreen interface for *iFire* to enable intuitive and interactive engagement within an immersive virtual environment. The tablet offered an alternative view of wildfire data, enhancing

context and interactivity while seamlessly integrating user inputs with the virtual environment [24, 25]. Users found the tablet intuitive and easy to use due to its familiar design and straightforward UI layout. Previous studies have highlighted the effectiveness of tablet-based tools in improving navigation and interaction within immersive environments, supporting learning and user engagement [17, 31, 47, 53]. **These findings suggest that integrating a touchscreen interface into immersive systems potentially improves user experience and interaction.**

Initial firefighter feedback highlighted the tablet's ability to provide accessible fire-related data, enhancing their understanding of wildfire scenarios. While we cannot claim that users experienced a fully visceral engagement without further analysis, combining a familiar touchscreen interface and immersive environments appears promising. Prior research indicates that immersive systems naturally elicit real-world reactions from users when realistic enough [40, 43], and interacting with familiar physical objects, such as tablets, can enhance a sense of presence and immersion [16]. These factors point to the potential for future refinements to foster deeper situational awareness and engagement during wildfire training in immersive environments.

6 Future Work and Conclusion

Future iterations will enhance functionality and user experience based on firefighter feedback and additional user research. Planned improvements include a landmarking tool for marking areas and gyroscope-based movement controls to simplify and potentially improve navigation. Collaboration with a partner fire institution

and further firefighter interviews will guide these enhancements, supporting persona development and validating design decisions. Additionally, findings highlight the importance of clear visual cues, intuitive interaction methods, and additional data overlays to improve situational awareness and decision-making. This iterative process aims to refine the interface and establish a foundation for creating more effective training modules, specially while integrating operational guidelines like LACES.

In conclusion, this paper has demonstrated the potential of using a tablet to explore an immersive environment for creating a compelling and engaging firefighter training system. By combining UCD and DT cycles, we have developed a prototype that meets firefighters' specific needs and provides a foundation for further improvements. The insights from user feedback are invaluable for refining the system and ensuring its effectiveness in real-world applications.

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A Personas

A.1 Persona 1

Dr Jack Napier is a 48-year-old fire scientist and firefighter. He has a deep passion for understanding and combating wildfires. He has dedicated his career to researching fire behavior. Jack’s commitment to advancing wildfire science led him to explore immersive data visualization technologies. His goal is to visualize and comprehend how variables impact the spread of wildfires, contributing to safer and more effective firefighting strategies.

Goals: Develop a comprehensive understanding of wildfires through data visualization. Create immersive training modules for experienced firefighters. Visualize and comprehend the impact of different variables on fire spread.

Challenges: Lack of immersive systems for exploring wildfire visualizations. Difficulty in finding tools that allow a detailed exploration and visualization of variables affecting fire spread.

Pain Points: Frustration with the current limitations in visualizing wildfire data. Inability to create effective training modules to combat wildfires in safe environments.

A.2 Persona 2

Commander Michael Garfield, a 35-year-old seasoned firefighter, led firefighting efforts during a massive wildfire incident in his home country. There, he witnessed the increasing challenges posed by wildfires. After that event, he began to seek innovative solutions for fire training. Faced with outdated training methods and limited exposure to real wildfire scenarios, he aims to improve understanding and modernize firefighting strategies. Frustrated by the lack of such tools in Australia, he aspires to leverage cutting-edge technology to create immersive training modules for experienced firefighters. He hopes this will lead to a revolutionized approach to developing planning and training methods.

Goals: Improve understanding of wildfires to enhance firefighting strategies. Plan and execute better approaches to combat wildfires. Modernize training methods for experienced firefighters.

Challenges: Limited exposure to real wildfire scenarios for training. Outdated training methods that don’t reflect current challenges.

Pain Points: Planning effective firefighting strategies is difficult due to an incomplete understanding of fire spread behavior. Concerns about the safety and relevance of current training methods.

A.3 Persona 3

Chuck Mathers, a 24-year-old aspiring firefighter, has always admired the bravery of local firefighters. However, the fear of the unknown and the desire to be adequately prepared for the challenges of future wildfires heavily weigh on his mind. He joined a firefighting academy with great expectations and enthusiasm, yet immediately identified the absence of a safe environment to gain hands-on experience in wildfire events. Eager to learn from experienced firefighters, Chuck envisions a future where realistic simulations help him overcome his fears and develop the skills to protect his community from the ever-present threat of wildfires.

Goals: Overcome fear and anxiety related to experiencing a wildfire. Learn from experienced firefighters in a safe and controlled environment. Acquire skills and knowledge necessary for effective firefighting.

Challenges: Fear of freaking out in the field during a wildfire. Limited exposure to real-world wildfire scenarios during training.

Pain Points: Anxiety about facing a wildfire without proper preparation. The desire for a safe and controlled environment for experiential learning.

B First Iteration

For the first iteration and based on early understanding of the users needs and indights, we designed the tablet interface (see Fig.3) to have two versions: **Aerial Mode View** and **Ground Mode View**. These elements change based on the perspective status.

The interface is divided into different sections. Each section in the interface groups features by themes that will adapt depending on the users' needs. Although elements inside the interface might change depending on the perspective mode, the themes of each section are intended to maintain familiarity.

The first section is *View Tools* (see Figure 3A, highlighted in green). These buttons allow firefighters to switch between **Aerial Mode View** and **Ground Mode View**. The second section is the *Global Navigation Information* (see Figure 3B, highlighted in blue). This section will change depending on the perspective mode and will constantly show the map of the entire fire-affected area. However, in the **Ground Mode View**, the map will be slightly compressed to show a simplified version of the ground view and provide the user with height information relative to the ground.

The third section is called the *Variable Bay* (see Figure 3C, highlighted in yellow). In this section, users can toggle on and off additional information on the variables that make up the wildfire. The fourth and final section is the *Space and Time Navigation* (see Figure 3D, highlighted in Purple). Here, we placed movement controls such as elevation, moving forward or backward, and the timeline to navigate the fire's progression.

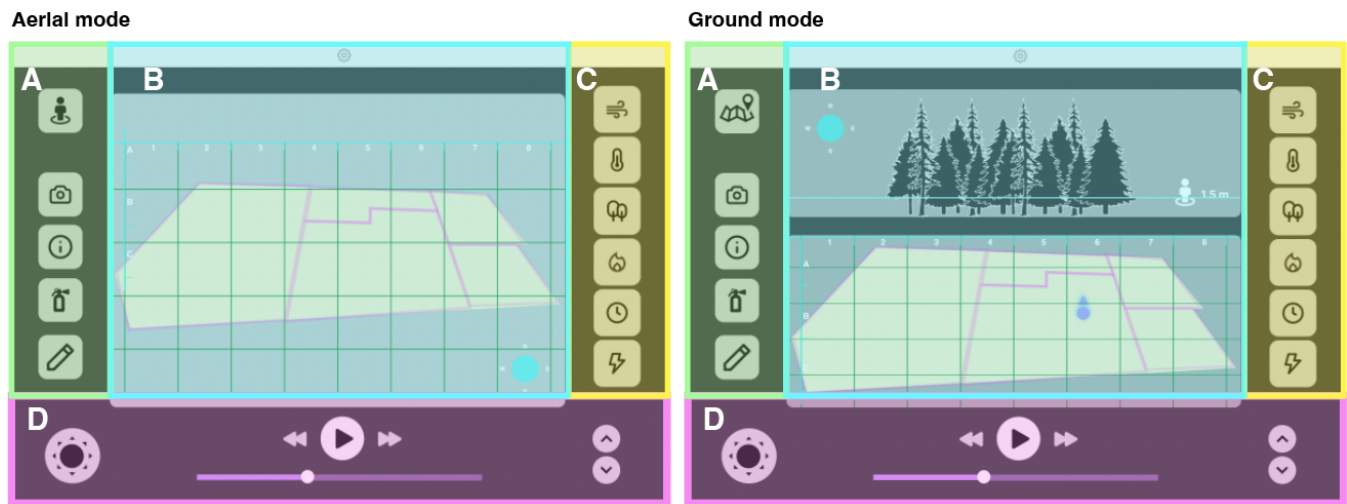


Figure 3: UI layout of both modes, aerial on the left and ground on the right.